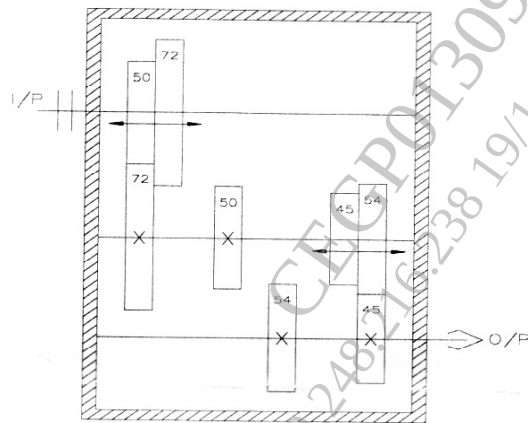


**MECHANICAL SYSTEM DESIGN
(2012 Course) (Semester - II) (402048)****Time : 3 Hours]****[Max. Marks : 70****Instructions to the candidates:**

- 1) **Answers Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.**
- 2) **Neat diagram must be drawn wherever necessary.**
- 3) **Figures to the right indicate full marks.**
- 4) **Use of programmable calculator is not allowed.**
- 5) **Assume suitable data wherever necessary.**

- Q1) a)** Figure 1 shows gearing diagram of a multispeed gearbox with number of teeth on each gear as specified. Speed of input shaft is 700 RPM. Answer the following questions based on the figure. **[4]**

**Figure.1**

- i) Draw correct speed diagram of the gearbox.
 - ii) Write correct structure formula of the gearbox.
- b) It has been observed from a sample of 200 bearing bushes that the internal diameters are normally distributed with a mean of 40.010 mm and a standard deviation of 0.005 mm. The upper and lower limits for the internal diameter are 40.02 and 40.00 mm respectively. Calculate the number of rejected bushes. **[6]**

Z	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4	3.0
Area	0.3413	0.3849	0.4192	0.4452	0.4641	0.4772	0.4861	0.4918	0.4987

OR**P.T.O.**

Q2) a) What is the need and of multi-speed gearbox in drive system of a machine tool? [4]

b) What do you understand by the term “Reliability and Margin of safety”, in the modern engineering design? [6]

Q3) a) Horizontal belt conveyor is to be used for transporting 500 metric ton of iron ore per hour at belt speed of 1.5 m/s. Mass density of iron ore is 1800 kg/m³. If surcharge factor for polyamide belt is 0.0725, determine the required effective belt width. [4]

b) Draw and explain screw take up arrangement in belt conveyor [6]

OR

Q4) Derive the expression for frictional force between the idler and idler roller pin for carrying and return run idlers. [10]

Q5) a) How pressure vessels are classified as per IS 2825. Explain your answer with reference to the following. [6]

- i) Operating conditions
- ii) Material of construction
- iii) Radiography requirement

b) A tube, with 50mm and 80mm as inner and outer diameters respectively, is reinforced by shrinking a jacket of outer diameter 100mm. The compound tube is to withstand an internal pressure of 40MPa. The shrinkage allowance is such that the maximum tangential stress in each tube has same magnitude. Calculate

- i) Shrinkage pressure and
- ii) The original dimensions of tubes

Assume $E = 207 \text{ KN/mm}^2$ [12]

OR

Q6) a) A cylindrical pressure vessel of 1250 mm in diameter and 20 mm in thickness is provided with a nozzle of 200 mm inner diameter and 15 mm thickness. The extension of nozzle inside the vessel is 15 mm. The corrosion allowance is 2 mm while the weld joint efficiency for shell as well as nozzle is 85%. The design pressure is 3MPa. The yield strength of material for the shell and nozzle is 200 N/mm². Determine whether or not reinforcing pad is required for the opening. If so determine the dimensions of the reinforcing pad made out of a plate of 14 mm thickness. [12]

b) What are the different types of vessel support? Explain any one type of support for vertical cylindrical pressure vessel with the help of diagram. [6]

Q7) a) Write a note on the materials used for manufacture of a crank shaft. [6]

b) Design a cast iron piston for a four-stroke diesel engine : [10]

Material of piston = FG200

Thermal conductivity of piston material = 46.6 W/m/°C

Factor of safety = 5

Cylinder bore = 250 mm

Length of stroke = 300 mm

Speed = 600 rpm

Indicated Mean Effective Pressure = 4 bar

Mechanical Efficiency = 80%

Maximum Gas Pressure = 4MPa

Fuel Consumption = 0.25 kg/kW/hr

Higher Calorific Value of fuel = 44000 kJ/kg

Assume that 5% of the total heat developed in the cylinder is transmitted by the piston. The temperature difference between the centre and the edge of the piston head is 220 °C.

- i) Calculate the thickness of the piston head by strength and thermal consideration and decide the thickness.
- ii) State whether the ribs are required. If so, calculate the number and thickness of piston ribs.
- iii) State whether a cup is required in the top of the piston head. If so, calculate the radius of the cup.

OR

Q8) The following data is given for the I section ($4t \times 5t$) connecting rod of a Diesel engine [16]

Cylinder bore = 85mm

Length of connecting rod = 350mm

Cross sectional area of connecting rod = $11t^2$

Where t is thickness of web and flanges of I section

Maximum gas pressure = 4MPa

Factor of safety against buckling failure = 5

(l/d) ratio for piston pin bearing = 1.5

(l/d) ratio for crank pin bearing = 1.25

Allowable bearing pressure for piston pin bearing = 13MPa

Allowable bearing pressure for crank pin bearing = 11 MPa

Length of stroke = 140mm

Mass of reciprocating parts = 1.5kg

Engine speed = 2000 rpm

Material of bolt = Chromium molybdenum steel

Yield strength of bolt material = 450 N/mm^2

Factor of safety for bolts = 5

Density of connecting rod = 7800 kg/m^3

Calculate

- Dimensions of the cross section of connecting rod at middle.
- Dimensions of small and big end bearings.
- Nominal diameter of bolts for the cap
- Magnitude of whipping stress

Q9) a) Explain with suitable example the method of optimum design for the case of redundant specifications? [6]

b) A shaft is to be used to transmit a torque of 2000 N-m. The required torsional stiffness of shaft is 100 N-m/degree, while the factor of safety based on yield strength is 2. Using the maximum shear stress theory, design the shaft with the objective of minimizing the weight, out of the following materials. Also find out diameter and length of shaft for the same. Assume combined shock and fatigue load factor for twisting moment as 1.25. [10]

Material	Weight Density, N/m ³	Modulus of Rigidity, N/mm ²	Tensile Yield Strength, N/mm ²
Chromium Steel	77×10^3	84×10^3	420
Plain Carbon Steel	76.5×10^3	84×10^3	230
Titanium Alloy	44×10^3	42×10^3	900
Magnesium Alloy	17.5×10^3	15×10^3	225

OR

Q10) a) Explain briefly the guidelines a designer has to follow to ensure safety in products. [6]

b) A tensile bar of circular cross section is subjected to the cyclic tensile force (F) which varies from zero to maximum. Determine the most significant objective of maximizing the energy absorption capacity using factor of safety (FS). Following limitations are to be used in the design due to space and assembly restriction. [10]

$$d \leq d_{\max} \text{ and}$$

$$L_{\min} \leq L \leq L_{\max}$$

